Graph Smile-Extrapolation

Filling dark nodes from lit neighbours by an OT-regularized Bayesian graph

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Abstract

This is the Vol-Fitter's headline differentiator: extrapolating sparse smile observations to a *full universe* of smiles, across expiries and assets, by propagating signal through a graph whose nodes are smiles (underlying, T). The mathematics is a Gaussian inverse problem with an optimal-transport-regularized prior on the *increments*, not the levels — so the default answer is "nothing changed from the prior", and lit calibrations force only the coherent, graph-plausible changes that explain them. We build the directed graph (a row-normalized kernel, its stationary distribution, reversible and directed Laplacians), assemble the increment precision $Q_\Delta = D_\kappa + \eta L_{\mathrm{dir}}^{\beta} + \lambda(A_\rho + \nu I)^{-1}$ from three beliefs (local smallness, directed prediction, low-energy unbalanced-OT flux), prove it is a proper Gaussian prior, and condition on the lit nodes in covariance form exploiting $n_{\mathrm{obs}} \ll N$. The production spine — transported prior, lit-calibration innovation, graph posterior increment, dark-node reconstruction, quote comparison — is described with its provenance hierarchy, data-derived precision tiers, and per-edge directed amplitude β. A six-node case file, solved by the production code, shows the whole mechanism on one page; and a temporal leave-one-out backtest measures the graph's *skill* over the transported-prior baseline: on 25 assets across three historical regimes ($\sim$47k held-out scores), neighbour-supported indexes gain +10 to +76 ATM vol bps and ETFs +3 to +7; fully-dark single names behind lit indexes/ETFs — the product case — gain +8 to +14 bps in the August-2024 spike and +3.8 to +7.2 fully out-of-sample in the October-2022 bear, with unbiased standardized residuals ($\bar{\zeta} \approx 0$). A case file documents the harness topology defect that had previously masked the cross-asset path entirely. A traceability table ties every claim to its module and test.

Contents

1 The production problem

The desk does not own one smile; it owns a universe of them. A handful of nodes — index expiries, a few mega-caps — are richly quoted (*lit*); the rest — single-name expiries, far tenors, the odd weekly that traded twice today — are barely quoted or not at all (*dark*). The dark nodes still need surfaces, and the wrong way to make them is to copy a nearby smile: copying creates jumps across expiry, ignores cross-asset information, and pretends that uncertainty did not grow with distance from the data. What production actually requires is the following.

Invariant

1. A dark node stays close to its transported prior unless lit data forces a move.
2. A move observed at one node propagates only through plausible graph paths: calendar neighbours first, cross-asset neighbours with smaller trust.
3. Every filled node reports an uncertainty, and uncertainty *widens* away from the lit observations.
4. The output is a smile-handle vector that retargets into the arbitrage-free smile machinery of Note 01 — the graph never emits a raw curve that could carry butterfly arbitrage.

The Vol-Fitter's answer treats the universe as a *graph*: nodes are smiles (underlying, T), edges encode "this node informs that one" (calendar edges within a name; cross-asset edges from an index or sector ETF to its single names). A prior surface ties the whole graph together; today's lit calibrations perturb it; the perturbation propagates.

Heuristic

The central modelling choice is to regularize the *increment* $z = x^1 - x^0$ — how each node moved from its transported prior — rather than the level x^1. The prior already encodes the shape of every smile; what the graph must infer is only the *change*, and changes are far more coherent across the graph than levels (an index move drags its names; a calendar shift is smooth in T). Regularizing increments makes "no change" the default answer, and data is required to *earn* every departure from the prior — exactly the right inductive bias for a sparsely observed universe.

The carrier is compact: each node propagates its three exact ATM handles $(\sigma_0, s_0, \kappa_0)$ of Note 01 — in code `atm_vol`, `skew`, `curvature` — as independent Gaussian coordinates (ATM *vol*, not total variance, so the level coordinate is comparable across expiries). Reconstruction retargets the posterior handles into an arbitrage-free smile. For exposition we write one scalar handle per node; production runs the same construction handle-by-handle with per-handle hyperparameters (section 5).

Figure 1: The production spine. The prior is transported first (Note 12). Lit calibrations produce genuine market innovations d = calibrated − transported. The graph posterior spreads those increments to dark nodes and carries uncertainty into reconstruction, which any residual dark quotes then score.

2 The graph object

Assumption 1. The user (or the auto-lattice) supplies raw weights $w_{ij} \geq 0$ on the node set $V = \{1, \ldots, N\}$, read "node j informs node i". Nodes with no out-neighbours receive a unit self-loop, so every row has positive mass.

Definition 1 (Kernel, stationary mass, conductance)**.** Row-normalization gives the row-stochastic kernel

$$K_{ij} = \frac{w_{ij}}{\sum_\ell w_{i\ell}}, \qquad \sum_j K_{ij} = 1.$$

A stationary distribution is a positive solution of $\pi^\mathsf{T} K = \pi^\mathsf{T}$, $\sum_i \pi_i = 1$; write $\Pi = \mathrm{diag}(\pi)$. The *reversibilized conductance* of the unordered pair $\{i, j\}$ is the symmetrized traffic

$$c_{ij} = \tfrac{1}{2}(\pi_i K_{ij} + \pi_j K_{ji}) = c_{ji} \geq 0.$$

Remark 1 (Reducible graphs)**.** Sparse user graphs can be reducible, and then the stationary solve $(K^\mathsf{T} - I)\pi = 0$ is singular. Production first attempts the direct dense solve; only on failure does it fall back to the uniform-teleport (PageRank) damping $K \mapsto (1-\delta)K + \delta/N$ with $\delta = 10^{-3}$, and as a last resort the uniform mass $1/N$. Irreducible graphs never enter the fallback, so their π is byte-identical to the undamped solve.

Two symmetric energies are built from these objects, and they answer different questions.

Definition 2 (Reversible Laplacian)**.** With $B \in \mathbb{R}^{N \times E}$ the signed incidence matrix of the conductance edges and $C = \mathrm{diag}(c_e)$,

$$L_{\mathrm{rev}} = BCB^\mathsf{T}, \qquad z^\mathsf{T} L_{\mathrm{rev}} z = \tfrac{1}{2} \sum_{i,j} c_{ij}(z_i - z_j)^2.$$

L_{rev} measures undirected *roughness*: it is PSD and annihilates constants. It never enters the prior directly; its role is to supply the geometry of the transport term (definition 4).

Definition 3 (β-weighted directed residual)**.** For a per-edge amplitude matrix β (with $\beta_{ii} = 1$), let $K^\beta_{ij} = K_{ij}\beta_{ij}$ (the Hadamard rescaling $K \circ B$). The *directed residual energy* of an increment field z is

$$\mathcal{E}_{\mathrm{dir}}(z) = \left\| (I - K^\beta) z \right\|_\Pi^2 = z^\mathsf{T} L^\beta_{\mathrm{dir}} z, \qquad L^\beta_{\mathrm{dir}} = (I - K^\beta)^\mathsf{T}\, \Pi\, (I - K^\beta).$$

The directed residual is the *prediction* rule: node i is compared with the K-weighted, β-scaled forecast $\sum_j K_{ij}\beta_{ij} z_j$ made by its in-neighbours, and the disagreement is charged at the node's stationary importance π_i. The kernel decides *who* predicts (K = trust); beta decides *how large* the transported increment should be along that edge (β = amplitude). A one-vol-point index ATM move may imply a 0.7-vol-point move in a single name while the skew handle translates at a different rate — so β is per-edge, per-direction and per-handle, and $\beta_{ij} \neq \beta_{ji}$ is allowed. Setting $\beta \equiv 1$ recovers the plain directed rule exactly (byte-identical, test-locked).

Proposition 1 (Directed residuals are safe Gaussian penalties)**.** *For every real amplitude matrix β and every π with $\pi_i > 0$, L_{dir}^{β} is symmetric positive semidefinite.*

Proof. $L_{\text{dir}}^{\beta} = R^{\mathsf{T}} \Pi R$ with $R = I - K^{\beta}$ is a Gram matrix in the Π-weighted inner product: for every z, $z^{\mathsf{T}} L_{\text{dir}}^{\beta} z = \sum_i \pi_i \big(z_i - \sum_j K_{ij} \beta_{ij} z_j\big)^2 \geq 0$, and symmetry is manifest from the factorization. □

Remark 2. Unlike L_{rev}, the directed residual does *not* annihilate constants unless every row of K^{β} sums to one: with $\beta \equiv 1$ a constant field is a perfect prediction ($L_{\text{dir}} \mathbf{1} = 0$), but an amplifying edge ($\beta > 1$) makes even a constant move partially “unexplained”. That is intentional — amplitude is a statement about how moves scale, not a reparametrization of trust.

The stationary solve and the two-line directed Laplacian are exactly the production computation (appendix C verifies the distilled listing against `volfit.graph` to 10^{-12}):

Listing 1: The stationary distribution and the β-weighted directed residual (distilled from `graph/{build,beta}.py`).

```
def stationary(K):                    # solve pi^T K = pi^T, sum(pi) = 1
    n = len(K); S = K.T - np.eye(n); S[-1] = 1.0  # last row = mass constraint
    rhs = np.zeros(n); rhs[-1] = 1.0
    return np.linalg.solve(S, rhs)

def directed_laplacian_beta(K, pi, B):   # L_dir^beta, PSD for ANY B (Prop 1)
    M = np.eye(len(K)) - K * B           # Hadamard rescale: trust x amplitude
    return M.T @ (pi[:, None] * M)
```

3 The increment prior

The increment field receives the Gaussian prior $z \sim \mathcal{N}(0, K_{\Delta})$, $K_{\Delta} = Q_{\Delta}^{-1}$, with precision

$$\boxed{Q_{\Delta} = D_{\kappa} + \eta\, L_{\text{dir}}^{\beta} + \lambda\, (A_{\rho} + \nu I)^{-1}.} \tag{1}$$

The three terms are three distinct answers to “what is a plausible change?”

Local smallness, $D_{\kappa} = \operatorname{diag}(\kappa)$, $\kappa_i > 0$.
A dark, poorly connected node must not invent a large move on its own. This is the cost of moving alone, and it is what makes the prior proper (proposition 3).

Directed prediction, $\eta\, L_{\text{dir}}^{\beta}$.
A node’s move should be explained by the nodes that inform it — the calendar and cross-asset propagation rule of definition 3.

Transport tangent norm, $\lambda\,(A_\rho + \nu I)^{-1}$.

A move is also plausible when it could have been *produced* by a coherent, low-energy flux along the edges, with net creation and destruction (sources and sinks) priced by ν. This is the tangent norm of unbalanced optimal transport on the graph [2, 1], and it asks a genuinely different question from L_{dir}: not “do my neighbours predict me?” but “could this pattern of changes have flowed here cheaply?”

Definition 4 (Mobility Laplacian)**.** For a positive reference density ρ on the nodes (default: the stationary mass), the *mobility* of edge $e = \{i, j\}$ is $m_e = c_e\,\theta(\rho_i, \rho_j)$ with θ the logarithmic mean $\theta(a, b) = (a-b)/(\log a - \log b)$ [3], and

$$A_\rho = B\ \mathrm{diag}(m_e)\,B^\mathsf{T}.$$

A_ρ is PSD and annihilates constants; it is L_{rev} with conductances reweighted by how much probability mass is locally available to move.

The transport term is easiest to read through its dual.

Proposition 2 (Screened-potential form of the OT term)**.** *For $A_\rho \succeq 0$ symmetric and $\nu > 0$,*

$$z^\mathsf{T}(A_\rho + \nu I)^{-1} z = \sup_{\phi\in\mathbb{R}^N} \Big\{ 2\langle z, \phi\rangle - \phi^\mathsf{T}(A_\rho + \nu I)\phi \Big\},$$

and the maximizer is the solution of the screened Poisson equation $(A_\rho + \nu I)\phi = z$.

Proof. The map $\phi \mapsto 2\langle z, \phi\rangle - \phi^\mathsf{T}(A_\rho + \nu I)\phi$ is strictly concave because $A_\rho + \nu I \succ 0$; its stationarity condition is $(A_\rho + \nu I)\phi = z$, and substituting the maximizer $\phi^\star = (A_\rho + \nu I)^{-1} z$ gives the stated value. □

So a graph increment is cheap under the third term exactly when the screened potential needed to explain it is small: smooth, transportable patterns need a gentle potential; an isolated spike at a badly connected node needs a large one. The allowance ν prices the sources — as ν grows, un-transported creation becomes cheaper and the term relaxes toward a scaled identity.

Proposition 3 (The increment prior is proper)**.** *Assume $\kappa_i > 0$ for all i, $\eta \geq 0$, $\lambda \geq 0$, $\nu > 0$, and $A_\rho \succeq 0$ symmetric. Then Q_Δ in (1) is symmetric positive definite, hence $z \sim \mathcal{N}(0, Q_\Delta^{-1})$ is a proper Gaussian law.*

Proof. $D_\kappa \succ 0$. $L^\beta_{\mathrm{dir}} \succeq 0$ by proposition 1. Since $A_\rho \succeq 0$ and $\nu > 0$, $A_\rho + \nu I \succ 0$, and the inverse of a positive definite matrix is positive definite. A nonnegative combination of PSD matrices added to the positive definite D_κ is positive definite; symmetry is preserved by each term (production additionally symmetrizes $\frac{1}{2}(Q + Q^\mathsf{T})$ against floating-point drift). □

> **Heuristic**
>
> The prior is a three-person committee. The first member says “do not move unless you must” (κ). The second says “if you move, the nodes that inform you should have predicted it” (η). The third says “and the whole pattern of moves should have been transportable along the graph without an implausibly expensive flow” (λ, ν). In the shipped default the third member abstains ($\lambda = 0$): the machinery is implemented, tested and exposed in the UI (“λ OT flux ($0 = $ off)”),

but the default product configuration runs the classical $D_\kappa + \eta L_{\mathrm{dir}}$ regime until the selected universe has enough topology for transportability to say something trust-worthy.

4 Conditioning on the lit nodes

Let $S = \{s_1, \ldots, s_m\}$ be the lit nodes, $H_S : \mathbb{R}^N \to \mathbb{R}^m$ the selector of their coordinates, and $d_S = x_S^1 - x_S^0$ the observed innovations with per-node observation precisions $r_s > 0$:

$$d_S = H_S z + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, R^{-1}), \quad R = \operatorname{diag}(r_s).$$

The baseline x^0 itself carries finite precision $p_i^0 > 0$ (the provenance tiers of section 5), which folds into the predictive covariance:

$$\mu^- = 0 \text{ (or a drift)}, \qquad K^- = K_\Delta + P_0^{-1}, \qquad P_0 = \operatorname{diag}(p^0).$$

Infinite trust in the baseline is the limit $P_0^{-1} \to 0$.

Proposition 4 (Covariance-form graph update)**.** *The posterior law of z given d_S is Gaussian with*

$$S_y = H_S K^- H_S^\intercal + R^{-1}, \tag{2}$$
$$\alpha = S_y^{-1}(d_S - H_S \mu^-), \tag{3}$$
$$\mu^+ = \mu^- + K^- H_S^\intercal \, \alpha, \tag{4}$$
$$K^+ = K^- - K^- H_S^\intercal \, S_y^{-1} \, H_S K^-. \tag{5}$$

Proof. The joint law of (z, d_S) is Gaussian with covariance

$$\begin{pmatrix} K^- & K^- H_S^\intercal \\ H_S K^- & H_S K^- H_S^\intercal + R^{-1} \end{pmatrix},$$

and (2)–(5) is the Schur-complement conditioning of the first block on the second [4]. □

Remark 3 (Why covariance form)**.** Production uses (2)–(5) rather than a posterior-precision assembly because $m = n_{\mathrm{obs}} \ll N$: only the m observed *columns* $K^- H_S^\intercal$ and one $m \times m$ solve are needed, and the reported marginals K_{ii}^+ come from a single `einsum` over those columns — no dense $N \times N$ posterior matrix is ever formed. A posterior variance ≤ 0 (numerically broken input) raises immediately rather than propagating a negative band.

Caution

The confidence reported for node i is $\pi_i^+ = 1/K_{ii}^+$ — the reciprocal of the *marginal* posterior variance — and never $(Q^+)_{ii}$, the diagonal of the posterior precision. The two differ whenever nodes are correlated (always): Q_{ii}^+ answers the conditional question "how pinned is i with all other nodes *held fixed*?", which overstates confidence precisely at the dark nodes whose neighbours are themselves uncertain. The accessor is `GraphPosterior.marginal_precision`, and the credible bands of fig. 2 use the marginal variance.

Figure 2 is a real run of the production solver on a calendar chain with one lit node: the unit innovation propagates outward and decays, the far node receiving only 15% of the lit move while the posterior standard deviation grows from 0.10 at the lit node to 1.52 at the far end — the credible band widening exactly as invariant 3 demands. Figure 3 shows the directed-smoothness weight η controlling the reach.

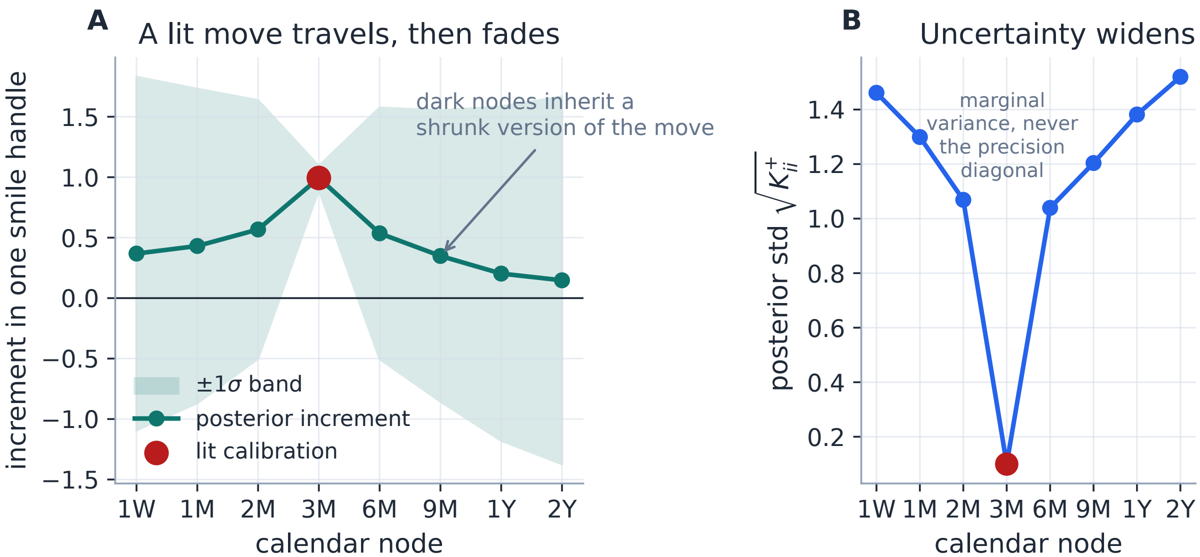

Figure 2: A one-handle innovation observed at the lit 3M node (red) propagates along a calendar chain and decays. The same update widens the uncertainty: the far node inherits 15% of the lit move while the posterior standard deviation grows from 0.10 to 1.52. Panel B is the reason the caution above exists: what is plotted is the *marginal* band width, which honestly decays away from the data.

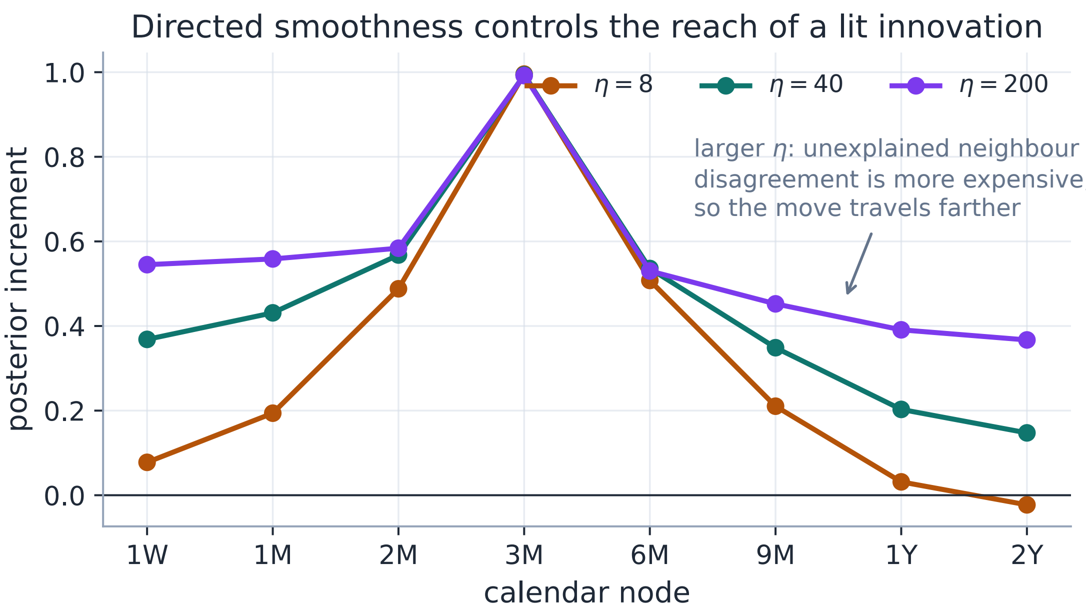

Figure 3: The directed-smoothness weight η controls reach. Small η leaves the move local (the κ member of the committee wins); large η makes unexplained neighbour disagreement expensive, so the move travels farther along the chain.

5 The production spine

5.1 Transported prior and provenance

Each node's baseline x^0 is the prior's three handles *transported to the current spot* (Note 12). The prior is resolved by a strict hierarchy, each level carrying its provenance tag and a validity flag: `active_transported` (a saved prior) $\rightarrow$ `nearest_expiry_transported` (precision scale 0.25) $\rightarrow$ `today_bootstrap` (0.05; not valid for validation — scoring against it would be circular) $\rightarrow$ `flat_atm` (0.01; diagnostic only). The provenance sets the baseline precision tier: a fragile fallback must never impersonate a saved prior.

5.2 Data-derived precision, not hand-set weights

Both precisions entering proposition 4 are *derived*:

$$r_s = \mathrm{clip}\Big(\tfrac{1}{\max(\mathrm{rms},\,\mathrm{floor})^2} \cdot f_{\mathrm{quotes}} \cdot f_{\mathrm{spread}} \cdot f_{\mathrm{age}} \cdot c_h,\ \cdot\Big), \qquad p_i^0 = \mathrm{clip}\Big(\mathrm{SOURCE} \cdot f_{\mathrm{age}} \cdot f_{\mathrm{transport}} \cdot c_h,\ \cdot\Big),$$

with $c_h =$ `HANDLE_CONFIDENCE` $= [1, 1, 0.01]$ (curvature is far harder to identify than level or skew) and per-handle floors and caps so that no single calibration dominates and no baseline vanishes. At the design point (active prior, dense fresh quotes, tight spread) the tiers reproduce the legacy regime $[10^6, 10^6, 10^4]$ exactly.

5.3 Lit innovation, not manual shift

For each lit node the observation is d = calibrated handles − transported prior handles: the node is actually fitted, in pure-market mode, so the innovation is the genuine market-versus-prior move and not itself prior-anchored. Dark nodes are never observations — a dark quote may be used *afterwards* for validation, never during propagation.

5.4 Per-handle hyperparameters

Production solves (1) once per handle with a per-handle regime (s_h, η_h): $\kappa = \texttt{kappaScale}/s_h^2$, $\eta = \eta_h \cdot \texttt{etaScale}$, $\lambda = \texttt{lambdaScale}/s_h^2$, where s_h is the handle's natural move scale (0.03 vol for ATM, 0.05 for skew, 0.5 for curvature) and η_h its base coupling (2×10^4, 7×10^3, 70). Scaling by $1/s_h^2$ keeps the *dimensionless* strength of each belief comparable across handles.

5.5 Amplitude: where β comes from

In the production API, β comes from the per-edge editor and the `crossBeta` scalar, entering only L^{β}_{dir} (never the mobility term). The temporal backtest harness additionally builds *vol-normalized* betas — $\beta_{\text{abs}} = \beta_{\text{vn}} \cdot \sigma_{\text{from}}/\sigma_{\text{to}}$ with $\beta_{\text{vn}} = \sqrt{T_{\text{to}}/T_{\text{from}}}$ on calendar edges and 0.7/0.8/0.6 for index/ETF/same-sector-name edges, capped at 3 — so that "a one-vol move over there" is translated into local vol units before amplitude is applied. That construction lives in the backtest edge builder; the core solver just consumes whatever β it is given.

5.6 Reconstruction and comparison

The posterior handles at a dark node are retargeted (Note 01's ATM-orthogonal chart, $w_0 = \sigma_0^2\tau$) into an exact arbitrage-free LQD slice — shape modes untouched, invariant 4 by construction. If the app is displaying SVI or Multi-Core SIV, the parametric family is refitted to the reconstructed target so the overlay matches the displayed model; for local volatility the reconstruction becomes the projection target of an affine LV calibration. The credible band rides along from the level handle's marginal variance. Where a dark node does own a few quotes, the reconstruction is scored against them — weighted RMS, inside-spread hit rate, and the standardized residual $\zeta = (\text{actual} - \text{posterior mean})/\sqrt{\text{posterior variance}}$ — an honest, never-trained-on measure.

6 A six-node case file

Case file: one index move, five listeners

Setup. The miniature universe of fig. 4: an SPX calendar chain (1M/3M/6M), the sector ETF XLK 3M, and two technology names AAPL/MSFT 3M. Calendar edges carry high trust both ways; XLK listens to SPX far more than SPX listens back; the names listen to their sector ETF first and the broad index second, with weak feedback edges keeping the chain irreducible. Lit nodes: SPX 3M calibrates +1.00 handle units above its transported prior, XLK 3M +0.55. The single names carry weaker baseline precision, so they are *allowed* to listen more.
Mechanism. The SPX calendar neighbours inherit most of the index move through the high-conductance calendar edges. AAPL and MSFT inherit a blend of the XLK and SPX innovations through their cross-asset edges, shrunk toward zero by κ and their own baselines.

Precision is highest at the lit nodes and decays along every path (solved by the production `build_graph/build_increment_prior/posterior_update`; the table is generated, not typed):

Node	Role	Posterior increment	Marginal precision
SPX 1M	dark	0.641	0.11
SPX 3M	lit	1.000	240.09
SPX 6M	dark	0.641	0.11
XLK 3M	lit	0.550	130.07
AAPL 3M	dark	0.435	0.05
MSFT 3M	dark	0.435	0.05

Lesson. The graph does not fill dark nodes by nearest-neighbour copying. Every dark node receives a *shrinkage forecast with a provenance path and a confidence*, and the confidence is what lets downstream quote comparison distinguish "the reconstruction is wrong" from "the reconstruction is far from lit information".

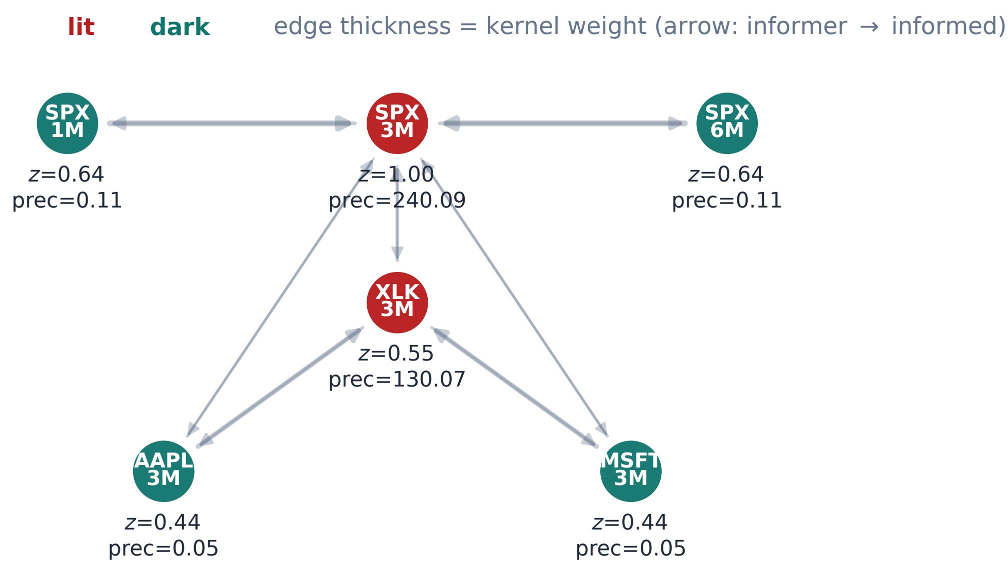

Figure 4: The six-node case file, solved by the production code. Red nodes are lit innovations; teal nodes are dark posterior reconstructions, each labelled with its posterior increment z and marginal precision. Arrow direction is informer → informed; thickness is the kernel weight after row normalization.

7 Does it work? The leave-one-out backtest

The graph's value is its *skill* over the transported-prior baseline: does propagating the lit innovations beat just keeping the (transported) prior? The temporal leave-one-out harness answers exactly that. Per consecutive captured day pair, freeze day-($T-1$) as the prior, transport it under the spot-dynamics regime R (Note 12), form the lit innovations, propagate, and compare the graph posterior at *held-out* nodes with their actual day-T calibration — against the pure transported-prior baseline.

Because the transport regime itself is uncertain, the harness brackets it: $R = 0$ (sticky-moneyness) leaves an underperformer's baseline unmoved and *over*-credits the graph; $R = 1$ (sticky-strike) bakes in the full leverage effect and *under*-credits it. The truth lies between.

On the spike regime (August 2024; 18 consecutive day pairs, 4134 scored held-out node evaluations):

- **Fill a sparse node from its lit neighbours** (withhold each clean node): the graph *decisively* beats transport — ATM skill +37.1 vol bps at $R = 0$ and +25.6 at $R = 1$, wing RMS skill +6.6 and +2.6 bps, with standardized residuals unbiased ($\bar{\zeta} = -0.01$) and slightly conservative ($\operatorname{std}\zeta = 0.90$ and 0.72) — the bands are, if anything, a touch too wide, which is the safe side of miscalibration. The driver is the calendar coupling.
- **Cross-asset to fully-dark names** (lit = index/ETF only): skill ≈ 0 on this pilot. Two explanations were recorded at the time — a pinning baseline precision (a 96 bp SPX innovation was measured to move a fully-dark AAPL node by 0.01 bp) and a structurally starved 8-asset universe. *Both later proved to be symptoms of a harness topology defect*, not properties of the method or the data; the case file below documents the defect, its fix, and the corrected verdict.

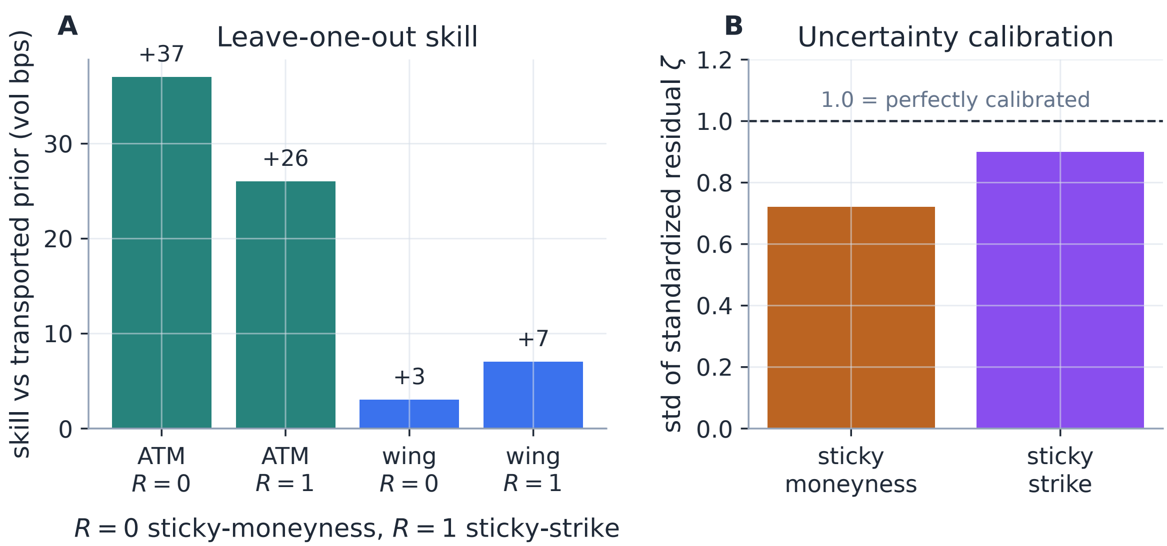

Figure 5: The leave-one-out verdict (8-asset pilot run; the 25-asset benchmark below ships as a regenerable HTML artifact). Left: skill over the transported prior for the neighbour-supported use case, bracketed by the two transport regimes — the graph adds +25.6 to +37.1 ATM vol bps. Right: the standardized-residual std is below one in both regimes: the posterior bands are slightly conservative, never overconfident.

7.1 Case file: the transient-name topology defect

The pilot's cross-asset null survived the move to the full 25-asset capture (3 regimes $\times$ 19 day pairs, $\sim$47k held-out scores): the liquid-split design returned ATM skill +0.000 in *every* regime — graph RMS equal to the baseline to three decimals — even though same-sector name–name and index$\to$name edges were live. Exact zeros are not damping; they are disconnection, and the diagnosis found it in the *harness*, not the solver. The backtest edge builder emitted cross-asset edges one-way (informer → name; indexes and ETFs received calendar edges only). Under the trust kernel K of definition 1 the single names were therefore *transient* states of the directed walk: all their mass drains into the index/ETF calendar chains and none returns, so the stationary distribution puts $\pi_i = 0$ on every name, the reversibilized conductance $c_{ij} = f(\pi, K)$ vanishes on every edge touching a name, and the increment prior decouples the dark names entirely. No signal ever arrived: the dark baseline precision never mattered (measured post-fix: a dead lever — identical posteriors at 0.25×, 0.05× and 0.01× the tier), and the "0.01 bp" pilot measurement was this artifact. The production auto-lattice is unaffected: its cross-ticker edges are symmetric, so $\pi > 0$ everywhere.

The fix (`EdgeConfig.cross_reverse_frac`, default 1) emits the reverse edge — the name informs its index/ETF informer — with the *inverse* amplitude β, so both directions encode the same linear relation and no second economic claim is introduced; names become recurrent and their conductance is nonzero. Setting the fraction to 0 reproduces the legacy disconnected topology (test-locked).

7.2 The 25-asset verdict on fully-dark names

With the topology fixed, the liquid-split design was re-run over all three regimes at reach $\eta = 10$ and cross-conductance ×25 — tuned on the spike regime only, so October 2022 and July 2023 are out-of-sample for the knob choice. Dark single names only (∼3.4–3.7k scores per regime and transport regime R; ATM, bps):

regime	R	graph RMS	baseline RMS	skill	ζ (mean/std)
spike Aug-2024	0	489.5	503.7	+14.2	−0.01 / 1.10
spike Aug-2024	1	475.5	483.5	+7.9	−0.06 / 1.02
high Oct-2022	0	193.6	200.9	+7.2	−0.02 / 0.78
high Oct-2022	1	174.3	178.1	+3.8	−0.03 / 0.70
low Jul-2023	0	451.6	452.3	+0.7	−0.04 / 1.91
low Jul-2023	1	452.0	452.8	+0.8	−0.06 / 1.85

Three conclusions. **(i)** Cross-asset extrapolation to fully-dark names carries real, quotable skill in stressed regimes: +7.9 to +14.2 bps in the spike and +3.8 to +7.2 fully out-of-sample in the October-2022 bear, with unbiased, honest-to-conservative bands. **(ii)** In the calm regime the skill is ≈ 0 — never negative — because single-name moves there are earnings-idiosyncratic and carry no systematic component the graph could transport; the mechanism cannot manufacture idiosyncratic information, only carry shared repricing. **(iii)** The one dishonest cell in the pack is the calm regime's dark-name bands (std $\zeta \approx 1.9$, overconfident): the dark posterior uncertainty should widen when a name's own regime is idiosyncratic (an event/earnings-aware term in the dark baseline precision) — an open follow-up.

8 What is original here

The ingredients are classical — Gaussian conditioning, graph Laplacians, stationary Markov chains, unbalanced-OT tangent norms. The synthesis for volatility is specific:

1. **Increments, not levels** (1): "no change from the transported prior" is the default dark-node answer, and lit data must earn departures.

2. **Three beliefs in one precision**: local smallness, directed prediction and transportable flux are separately tunable in a single proper Gaussian prior (proposition 3).
3. **Trust and amplitude separated**: K chooses the neighbours, β scales their increments, per edge, per direction and per handle (proposition 1 keeps every choice safe).
4. **Smile handles as the carrier**: the propagated coordinates are the exact ATM handles of Note 01, so every reconstruction retargets into an arbitrage-free smile rather than a free-floating scalar field.
5. **Skill, honestly measured**: the leave-one-out harness scores against the transported prior — the strongest cheap competitor — bracketed by the two stickiness regimes, with calibrated standardized residuals, rather than against a straw-man flat fill.

To our knowledge, propagating arbitrage-free smile handles through an OT-regularized Bayesian graph to reconstruct an unquoted universe is novel.

9 Traceability

Table 1: Claims in this note and the code/tests that lock them.

Claim	Object	Code anchor	Test anchor
Kernel, π, conductance; teleport fallback only when reducible	definition 1 and remark 1	`volfit/graph/build.py`	`test_graph_example.py`
Operators PSD, annihilate constants	definitions 2 and 4	`volfit/graph/operators.py`	`test_graph_example.py::test_operators_are_psd_and_annihilate_constants`
β-residual PSD for any β; $\beta \equiv 1$ byte-identical	proposition 1	`volfit/graph/beta.py`	`test_graph_beta.py`
Increment precision matches (1); proper prior	proposition 3	`volfit/graph/prior.py`	`test_graph_example.py::test_increment_precision_matches_note`
Covariance-form update; innovation system	proposition 4	`volfit/graph/posterior.py`	`test_graph_example.py::test_innovation_system_matches_note`
Reported confidence is marginal, $1/K^{+}_{ii}$	section 4	`GraphPosterior.marginal_precision`	`test_graph_example.py::test_posterior_table_matches_note`
OT term implemented; pulls toward baseline	proposition 2	`volfit/graph/operators.py`, `prior.py`	`test_graph_example.py::test_ot_weight_pulls_toward_baseline`
Provenance / precision tiers	section 5	`volfit/graph/precision.py`, `api/graph_nodes.py`	`test_graph_precision.py`, `test_graph_node_priors.py`
Lit innovation = calibrated − prior	section 5	`api/graph_extrapolation.py`	`test_graph_extrapolate_solve.py`

Claim	Object	Code anchor	Test anchor
Reconstruction is arbitrage-free	invariant 4	`api/graph_reconstruct.py`	`test_graph_reconstruct.py`, `test_graph_lv.py`
Held-out nodes predicted, not pinned	section 7	`api/graph_backtest.py`	`test_graph_backtest.py`
Directed vol-normalized back-test edges	section 5	`backend/backtest/graph_edges.py`	`test_graph_loo_backtest.py`
Reverse cross edge keeps names recurrent ($\pi > 0$); fraction 0 = legacy disconnected topology	section 7.1	`EdgeConfig.cross_reverse_frac`	`test_graph_loo_backtest.py::test_index_to_name_direction_and_vol_normalization`

A Hyperparameter atlas

Table 2: Graph-extrapolation hyperparameters.

Knob	Default	Role
Surfaced		
`graphKappaScale`	1.0	Scales the local precision $\kappa = \texttt{kappaScale}/s_h^2$; larger keeps nodes closer to the prior.
`graphEtaScale`	1.0	Scales the directed-prediction weight $\eta = \eta_h \cdot$ `etaScale` (autotunable, appendix B).
`graphLambdaScale`	0.0	OT tangent-norm weight (off by default; "λ OT flux").
`graphNu` ν	0.1	Source/sink allowance in the screened operator $A_\rho + \nu I$.
`crossBeta` / `edgeBetas`	1.0 / —	Per-edge directed increment amplitude (definition 3).
graph `calendarWeight` / `crossWeight`	10.0 / 2.0	Auto-lattice edge trusts (calendar vs cross-ticker). *Not* the LQD calendar-slack penalty of Note 10, which shares the name `calendarWeight` in `OptionsSettings`.
`flatAtm`	`false`	Allow the flat-ATM diagnostic baseline.
Hidden		
`GRAPH_PRIOR_HYPER`	(s_h, η_h)	Per-handle scale and base coupling: $(0.03, 2{\cdot}10^4)$ ATM, $(0.05, 7{\cdot}10^3)$ skew, $(0.5, 70)$ curvature.
`HANDLE_CONFIDENCE`	$[1, 1, 0.01]$	Per-handle relative confidence (curvature is weakly identified).
`SOURCE_BASE`	tiered	Provenance precision: active $10^6 \gg$ nearest $2.5{\cdot}10^5 \gg$ bootstrap $5{\cdot}10^4 \gg$ flat 10^4.
`OBS/BASE_PRECISION` floors/caps	per-handle	No single fit dominates; no baseline vanishes.

Knob	Default	Role
`STATIONARY_DAMPING`	10^{-3}	Teleport damping, reducible graphs only (remark 1).
`sink_self_loop`	1.0	Self-loop weight for out-degree-zero rows.

B Performance notes

1. **Covariance-form conditioning.** Only the n_{obs} observed columns of K^- and one $n_{\text{obs}} \times n_{\text{obs}}$ solve are formed; the marginal variances come from one `einsum` over those columns. Cheap whenever few nodes are lit — the production case.

2. **Dense at $O(10^3)$.** Q_Δ and $K_\Delta = Q_\Delta^{-1}$ are formed densely, fine up to $\sim 10^3$ selected nodes. The deferred Phase-10 sparse path replaces the two dense $N \times N$ inverses with sparse solves $(A_\rho + \nu I)u = v$ and a Hutchinson diagonal estimator for universes $\gg 10^3$.

3. $\beta \equiv 1$ **golden.** The per-edge beta machinery is byte-identical to the plain directed rule at $\beta = 1$ (test-locked), so it is strictly additive.

4. η **autotune.** `POST /graph/autotune` chooses `etaScale` by leave-one-out cross-validation over the seven-point grid $\{0.1, 0.25, 0.5, 1, 2, 4, 10\}$ — the $O(7\, n_{\text{obs}} N^3)$ cost the sparse path targets.

C Reference implementation

The prior and the posterior update are each a handful of lines. The listing reproduces the production `build_increment_prior` and `posterior_update` (with $\lambda = 0$) to better than 10^{-10} on the six-node case file of section 6, including the marginal precisions printed in the table.

Listing 2: Increment prior and covariance-form update (distilled from `graph/{prior,posterior}.py`).

```python
def increment_prior(K, pi, kappa, eta, B=None):
    B = np.ones_like(K) if B is None else B
    L = directed_laplacian_beta(K, pi, B)
    Q = np.diag(np.full(len(K), kappa)) + eta * L
    return np.linalg.inv(0.5 * (Q + Q.T))   # K_Delta

def posterior(K_delta, base_prec, obs, y, r):
    k_minus = K_delta + np.diag(1.0 / base_prec)  # + P0^{-1}
    cols = k_minus[:, obs]                   # observed columns only
    S_y = cols[obs, :] + np.diag(1.0 / r)  # H K^- H^T + R^{-1}
    alpha = np.linalg.solve(S_y, y)         # innovation weights
    mean = cols @ alpha                     # mu^+  (mu^- = 0)
    var = np.diag(k_minus) - np.einsum(     # marginal K^+_{ii}:
        "ij,ji->i", cols, np.linalg.solve(S_y, cols.T))
    return mean, 1.0 / var                # confidence = 1/K^+_{ii}
```

References

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